



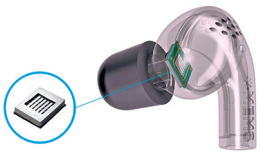
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

MEMS speaker delivers powerful low frequency sound

The Cypress solid-state MEMS speaker from xMEMS Labs is the industry's first to achieve a low-frequency sound pressure level (SPL) greater than 140dB. The company asserts that this speaker



represents a departure from conventional push-air sound methods

towards ultrasonic amplitude modulation transduction. It offers a new alternative to the moving-coil method, ensuring quality in active noise-cancelling (ANC) earbud micro speakers. The sound envelope of the ultrasonic pulse precisely replicates the original signal, making Cypress more faithful to the original recording across all frequencies compared to existing speaker technologies. In comparison to previous xMEMS speakers, it is 40 times louder in these frequencies, providing robust, deep bass similar to the best 10-12mm legacy coil speakers.

xMEMS Labs

<https://xmems.com>

Wi-Fi IC enhances location accuracy and efficiency

Nordic Semiconductor introduces the nRF7000 Wi-Fi companion IC, claiming to be the first to offer a complete silicone-to-cloud locationing solution incorporating Wi-Fi, cellular IoT, and GNSS. When used with Nordic's nRF91 series cellular IoT system-in-package (SiP), it enables SSID-based Wi-Fi locationing, highly power-efficient for accurate indoor and outdoor loca-

tion fixes, even in urban and suburban areas. The method determines a tracker's location by scanning signals from nearby Wi-Fi access points, rather



than relying on traditional cell towers or satellite signals. This approach provides location accuracy surpassing cellular-

based methods, though it is slightly less precise than GNSS. However, it offers greater power efficiency than GNSS and is nearly as efficient as cellular-based locationing.

Nordic Semiconductor

<https://www.nordicsemi.com>

MCU series for smart and industrial connectivity

The GigaDevice GD32VW553 series MCU supports Wi-Fi 6 and Bluetooth LE 5.2. Developed on a process platform, it caters to market applications requiring



wireless capabilities, such as smart home appliances, industrial internet, and communication gateways, providing edge

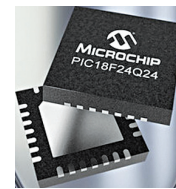
processing and connectivity. Additionally, it is suitable for office equipment, payment terminals, and IoT products. Supporting OFDMA allows multiple devices to share channel resources, increasing data transmission rates over Wi-Fi 4. The series also features a wireless coexistence protocol based on the PTA mechanism, minimising Wi-Fi and Bluetooth interference. Operating on a power supply voltage of 1.8V to 3.6V and tolerating temperatures up to 105°C, it is suitable for industrial control, lighting equipment, and socket panels.

GigaDevice

<https://www.gigadevice.com>

Secure, versatile microcontrollers for embedded systems

Microchip Technology introduces the PIC18-Q24 family of microcontrollers (MCUs). These MCUs include the Programming and Debugging Interface Disable (PDID) to address the risk of device reprogramming in embedded systems. When activated, PDID protects the programming/debugging interface,



preventing unauthorised reading, modification, or erasure of firmware. The MCUs also feature multi-voltage I/O (MVIO) to

accommodate interaction with various sensors, memory chips, and processors. MVIO allows the MCUs to interface with digital inputs or outputs at different voltages, eliminating the need for external level shifters. This function streamlines board design and reduces the bill of materials cost. MVIO also makes the MCUs suitable for system management roles, such as handling monitoring and telemetry in larger processors—essential tasks for security, as hackers often target them in embedded systems.

Microchip Technology

<https://www.microchip.com>

Ultra-wideband chip for versatile automotive uses

NXPs Semiconductors releases the Trimimension™ NCJ29D6, an automotive ultra-wideband (UWB) single-chip solution. This chip is intended for various applications, including secure car access, child presence detection, intrusion alert, and gesture recognition. The company claims it is the first automotive device to combine ranging